

High Capacity Spectrum Allocation for Multiple D2D Users Reusing Downlink Spectrum in LTE

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Abstract—Device-to-Device (D2D) communications is regarded as an option to extricate cellular Base Stations (BSs) from heavy traffic load. With more devices using D2D communications, there is an urgent demand for more spectrum bands for D2D communications. To provide more licensed bands for D2D communications, downlink spectrum reuse is a promising way to solve this problem. In an FDD-OFDMA LTE network, discontinuous resource blocks (RBs) of downlink spectrum are allocated to the same cellular user, thus resulting in complex interference between D2D users and cellular users. In order to increase the number of D2D users in the system and to increase the capacity of D2D users, allowing RBs reuse by more D2D pairs is an effective method, which, however, may cause more complicated interference. Multiple D2D users communicate with the same RBs poses a trade-off between transmission power and D2D users data rates. To provide a high capacity and to serve more users, we propose the RICE algorithm to allocate continuous RBs to D2D pairs, by facilitating different D2D pairs to share the same RBs. Simulations show that the RICE algorithm provides the highest capacity as compared to the exclusive spectrum allocation (ESA) strategy and adaptive subcarrier allocation (ASA) algorithm.

I. INTRODUCTION

With the rapid growth of smart interconnected devices, more devices are connected to the Internet [1], [2], thus posing great challenges to network providers, especially to those that serve wireless devices, such as cellular network providers. In a traditional cellular network, all data are gathered by base stations (BSs) and then transmitted to each user equipment (UE) [3], [4]. Currently, transmitting enormous amount of data by users through the cellular network requires more spectrum resources. Service providers are employing D2D communications to increase the total capacity without purchasing more spectrum resources. By using D2D communications in a cellular network, BSs can be potentially alleviated from the congestions and can provide a higher capacity for the system [5], [6]. Cellular users do not need to transfer all the data through BSs. Instead, after authorized by BSs, cellular users can transmit data directly to receivers in the form of D2D communications.

In strategizing D2D communications, spectrum reuse, where D2D users reuse the spectrum bands of cellular users, is an efficient way to increase the spectrum utilization, but at the expense of the service quality for cellular users. There are mainly two types of D2D spectrum reuse. One is to deprive

cellular users of some bands solely for D2D communications. The other is to let D2D users directly reuse the same spectrum bands of cellular users. Both of these strategies can provision D2D communications with licensed bands, while the second strategy has less impact on cellular users.

In a system with both cellular users and D2D users, and with D2D users reusing the spectrum of cellular users, the main factor that influences the capacity of D2D users is the interference. The interference mode varies with different spectrum reused by D2D users. In the uplink spectrum reuse strategy, all the interference from D2D users goes to BSs, and this increases the difficulty in differentiating signals of cellular users from noise and interference [7]. On the contrary, reusing downlink spectrum of cellular users may not affect the BS, but the interference received by cellular users increases. In order to alleviate BSs, reusing the downlink spectrum of cellular users is a better choice [8].

Using D2D communications in a cellular network is to provide a high capacity for D2D users, while maintaining the service quality for cellular users. In other words, the capacity of D2D users needs to be maximized while reusing the downlink spectrum of cellular users. Based on the constraints of UEs, D2D users can only use Single-Carrier FDMA (SC-FDMA), which only allows for one carrier per user. Therefore, D2D users need to be allocated RBs with continuous spectrum [9]. In order to maximize the capacity of D2D users, we introduce the laRgest flrst Continuous sElective (RICE) algorithm that reuses the downlink spectrum of cellular users. The main contributions are listed as follows: 1. We propose a system in which D2D users reuse the downlink spectrum of cellular users. Different D2D users can reuse the same RB in the system. 2. We formulate the problem of maximizing the total capacity of D2D pairs. 3. We design a heuristic algorithm, RICE, to maximize the total capacity of D2D pairs. The performance of RICE is demonstrated via extensive simulations.

The rest of the paper is organized as follows. In Section II, we briefly review the related works. In Section III, we propose a system in which multiple D2D users can reuse the same spectrum bands of cellular users, and present the mathematical analysis of the interference in the system. In Section IV, we formulate the problem of spectrum allocation for D2D users

to maximize the capacity of all D2D users. In Section V, we demonstrate the performance of the proposed algorithm via extensive simulations. Conclusions are presented in Section VI.

II. RELATED WORKS

Some of the existing studies on D2D communications focus on D2D users in reusing the uplink spectrum of cellular users. Since cellular users only transmit data to BSs in uplink spectrum, D2D communications in reusing uplink spectrum cannot interfere cellular users. Panigrahi *et al.* [10] used a two-phase uplink scheduling to maximize the capacity. Yang *et al.* [11] used a modified genetic algorithm, which enables one RB shared by one cellular user and two D2D pairs in uplink spectrum. Wang *et al.* [12] came up with a simple resource allocation strategy, which allows one D2D pair reuse RBs of one cellular user in uplink spectrum. Bao *et al.* [13] achieved spectrum allocation of uplink spectrum by using fractional frequency reuse strategy. Liu *et al.* [14] considered the green energy in D2D communications.

Although reusing uplink spectrum of cellular users has attracted great attention, the significant interference at BSs remains a major challenge yet to be addressed. As compared to reusing the uplink spectrum, reusing the downlink spectrum of cellular users does not cause great interference to BSs. Seif *et al.* [15] used successive interference cancellation to increase the performance of downlink spectrum reuse. Saha *et al.* [16] used the rate coverage probability to achieve the maximum rate coverage. Liu *et al.* [17] considered the distance between UEs with stochastic geometry distribution in utilizing dedicated spectrum bands for D2D users in downlink licensed bands.

These downlink spectrum resource allocation strategies in the aforementioned works do not consider multiple D2D pairs in reusing one RB with cellular users, which in fact can support more D2D pairs and provide a higher capacity of D2D pairs. By using the downlink spectrum for D2D users, there will be more choices of spectrum bands for D2D communications. Although using the downlink spectrum of cellular users imposes more complicated interference between UEs, it can still be solved by reasonable spectrum allocations. A system that allows different D2D users to reuse the same spectrum bands, especially the downlink spectrum bands of cellular users, poses three major challenges [18]. First, the allocation of RBs for D2D users needs to be spectrum continuous. In the downlink reuse strategy, the RBs allocated to cellular users are discontinuous in spectrum, implying that allocating RBs to one D2D user involves several cellular users. The second challenge is the interference between different D2D users. When one RB is allocated to many D2D users, these D2D users may interfere each other. It is necessary to assess the importance between transmission power and interference within D2D users. In addition, the required transmission rate of cellular users must be satisfied.

III. SYSTEM MODEL

For the D2D-assisted downlink OFDMA-based cellular network, there are two types of users, cellular users and D2D users. For each D2D transmitter, there is a paired D2D receiver. The uplink data transmission scheme, Single-Carrier FDMA, is adopted for UEs, *i.e.*, D2D transmitters can only use continuous spectrum resources. On the other hand, cellular users can receive data from OFDMA-based BSs by using discontinuous spectrum resources.

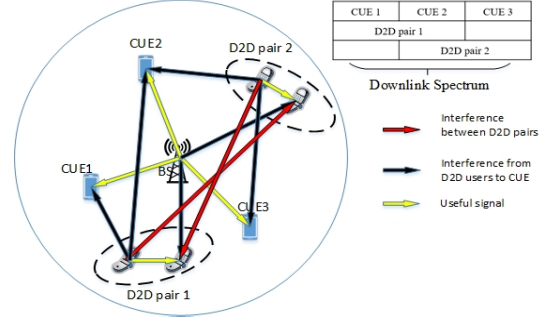


Fig. 1. Cellular network with D2D users reusing downlink spectrum bands.

Since licensed spectrum is valuable and limited, D2D pairs have to rely on the cellular network. In particular, D2D pairs may reuse the downlink spectrum of cellular users. In our system model, we assume that each base station collects all the channel state information and correspondingly allocates the spectrum resources to both downlink cellular users and D2D transmitters. The interference between cellular user equipments (CUEs) and D2D pairs, which is caused by spectrum reuse, is illustrated in Fig. 1. When the spectrum is assigned to one D2D pair, all involved CUEs (such as D2D pair 1 vs. CUE 1 and CUE 2) using this spectrum band receives interference from this D2D transmitter. If two or more D2D pairs share the same spectrum (such as D2D pair 1 vs. D2D pair 2), the receiver of each D2D pair may receive interference from all other involved D2D transmitters.

Assuming that cellular users have higher priority than D2D pairs, the spectrum resources allocated to cellular users should satisfy the user-specific data rate requirements. Since in the cellular network, to avoid intra-cell interference, one RB can only be reused by one cellular user, and cellular users can be represented by their assigned RBs.

According to the Shannon Formula, the relationship between the required data rate for RB v (R_v) and acceptable SINR value for RB v (ξ_v) is

$$R_v \leq \lambda \cdot \log_2 (1 + \xi_v), \quad (1)$$

where λ is the bandwidth of one RB. Thus, $\xi_v \geq 2^{R_v/\lambda} - 1$.

$$\xi_v = \frac{P_v \cdot h_v}{N_v + I_v}, \quad (2)$$

where P_v – transmission power of BS at RB v ;

N_v – noise at RB v ;
 I_v – interference received by the cellular user at RB v ;
 h_v – channel gain from BS to cellular user at RB v .

By plugging (2) into (1), the interference of RB v is $I_v \leq \frac{P_v \cdot h_v}{2^{R_v/\lambda} - 1} - N_v$. The maximum interference that can be tolerated at RB v ($I_v^{\max}$) is defined in Eq. (3)

$$I_v^{\max} = \frac{P_v \cdot h_v}{2^{R_v/\lambda} - 1} - N_v. \quad (3)$$

To ensure the data rate of the cellular user at RB v , the total interference at RB v should be less than $I_v^{\max}$, i.e.,

$$\sum_{j \in J_v} P_{v,j}^D \cdot h_{v,j}^D \leq I_v^{\max}, \quad (4)$$

where $P_{v,j}^D$ – transmission power of D2D pair j 's transmitter at RB v ;

J_v – the set of D2D pairs at RB v ;

$h_{v,j}^D$ – channel gain from the D2D pair j 's transmitter to the cellular user at RB v .

D2D pairs receive interference from two parts, BS and other D2D pairs using the same RBs. In the system, the SINR of D2D pair j at RB v ($\xi_{v,j}^D$) is

$$\xi_{v,j}^D = \frac{P_{v,j}^D \cdot h_{v,j}^p}{N_v + P_v \cdot h_{v,j}^C + I_{v,j}^D}, \quad (5)$$

where $h_{v,j}^p$ – channel gain of the D2D pair j 's transmitter to the receiver at RB v ;

$h_{v,j}^C$ – channel gain from the BS at RB v to D2D pair j ;

$I_{v,j}^D$ – the interference at the D2D pair j 's receiver from other D2D pairs at RB v .

The interference at D2D pair j 's receiver from other D2D pairs at RB v ($I_{v,j}^D$) is

$$I_{v,j}^D = \sum_{k \neq j, k \in J_v} P_{v,k}^D \cdot h_{k,j}^D, \quad (6)$$

where $h_{k,j}^D$ is the channel gain from the transmitter of D2D pair k to the receiver of D2D pair j .

For D2D pair j , $j \in J_v$, the capacity at this RB ($C_{v,j}^D$) is

$$C_{v,j}^D = \lambda \cdot \log_2 (1 + \xi_{v,j}^D). \quad (7)$$

IV. PROBLEM FORMULATION

The main purpose of D2D communications is to provide a high capacity of D2D users. Thus, we formulate the problem as follows:

$$\mathbf{P0} : \text{Max} \sum_{P_{v,j}^D} \sum_{J_v} \sum_{v \in V} \sum_{j \in J_v} C_{v,j}^D \quad (8)$$

$$\text{s.t.} : \xi_{v,j}^D \geq \xi_{v,j}^{D,th}, \quad \forall j \in J, \quad \forall v \in V, \quad (9)$$

$$\sum_{j \in J_v} P_{v,j}^D \cdot h_{v,j}^D \leq I_v^{\max}, \quad \forall v \in V, \quad (10)$$

$$a_{1,j} + a_{m,j} + \sum_{v=1}^{m-1} |a_{v+1,j} - a_{v,j}| = 2, \quad \forall j \in J \quad (11)$$

where the total number of RBs in set K is m . J and V denote the set of all D2D pairs and RBs, respectively. $a_{v,j}$ is the binary parameter of RB reuse. If RB v is reused by D2D pair j , $a_{v,j} = 1$; otherwise, $a_{v,j} = 0$. The objective is to maximize the total capacity of all D2D pairs. Constraint (9) is to guarantee that the SINR of D2D users is greater than its threshold in order to satisfy the receivers' sensitivity. $\xi_{v,j}^{D,th}$ is the SINR threshold of D2D pair j that reuses RB v . Constraint (10) is to guarantee that the interference from cellular users is within the tolerance. Constraint (11) is to ensure that the allocated RBs are spectrum continuous for each D2D pair.

Theorem1: Problem **P0** is an NP-hard problem.

Proof: If a special case of **P0** is an NP-hard problem, it can therefore be proved that **P0** is an NP-hard problem. For the special case, only one RB is considered, and the index of v is ignored. In addition, no continuous RB allocation exists, thus implying that Constraint (11) can be relaxed. The original problem can therefore be reduced to a special case **P1**

$$\mathbf{P1} : \sum_{j \in J_1} C_j^D \quad (12)$$

$$\text{s.t.} : \xi_j^D \geq \xi_j^{D,th}, \quad \forall j \in J, \quad (13)$$

$$\sum_{j \in J_1} P_j^D \cdot h_j^D \leq I^{\max}. \quad (14)$$

In order to prove **P1** to be NP-hard, we demonstrate that the decision problem of **P1** is NP-complete. The decision problem of **P1** can be described as follows: Given a set J , each item in J has a positive value C_j^D . Is it possible to find a feasible set J_1 that can maximize $\sum_{j \in J_1} C_j^D$ with Constraint (13) and (14) satisfied? The decision problem of **P1** is a quadratic knapsack problem (QKP), which is a well-known NP-complete problem. So, **P1** is an NP-hard problem, and thus **P0** is NP-hard.

V. LARGEST FIRST CONTINUOUS SELECTIVE ALGORITHM

The problem of spectrum allocation to maximize the capacity of multiple D2D pairs reusing RBs is an NP-hard problem. In order to solve this problem, a heuristic algorithm, the laRgest flrst Continuous sElective (RICE) algorithm, is proposed. The basic idea of RICE is to iteratively select the largest total capacity based on RB allocation in the previous iteration, and then extend the RB allocation.

Before running the simulations using the RICE algorithm, the system is initialized with parameters calculated by one D2D pair occupying one RB exclusively, and the initial capacity of D2D pairs at each RB is calculated. $m_{v,j}$ is a binary parameter in evaluating $I_v^{\max}$ and the interference received by cellular user at RB v ($I_{v,j}$). If $I_{v,j} < I_v^{\max}$, $m_{v,j} = 1$. Otherwise, $m_{v,j} = 0$.

The RICE algorithm is summarized in Algorithm 1. First, the D2D pair with the largest capacity at each RB is selected (Steps 3-10). At each RB, if $m_{v,j}$ for the selected D2D pair is 1, it means that this RB can be shared by other D2D pairs. The new D2D pairs are chosen so that all D2D pairs have the maximum total capacity and the total interference received by

cellular user at this RB is no more than I_v^{max} . If $m_{v,j}$ is 0, then this RB is exclusively occupied by the D2D pair with the largest capacity. The next step is to set the continuous RBs to D2D pairs, so as to ensure the highest capacity of each D2D pair (Steps 11-13). Then, $C_{v,j}^D$ is renewed based on the current RB allocation (Steps 14-19). The algorithm ends when there is no more RB can be reused. The output is set as J_v , which stores the final RB allocation.

Algorithm 1 RICE algorithm

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1: Input:  $C_{v,j}^D, H, V, J$ 
2: while  $\sum_{v \in V} \sum_{j \in J} m_{v,j} \neq 0$  do
3:   for all  $v$  in  $V$  do
4:     Find the largest  $C_{v,j}^D$  D2D pair  $\rightarrow J_v$ 
5:     if  $m_{v,j} = 1, j \in J_v$  then
6:       Renew  $C_{v,j}^D$  by  $P_{v,j}^D \leq (I_v^{max} - I_{v,j'})/h_{v,j}^D$ ,
         $j \in J, j \notin J_v$ 
7:       Find D2D pair with the largest  $C_{v,j}^D + \sum_{j \in J_v} C_{v,j}^D$ 
         $\rightarrow J_v$ 
8:       Renew  $m_{v,j}, j \in J$ 
9:     end if
10:  end for
11:  for  $j \in J$  do
12:    Find continuous RBs with the largest  $\sum_{v \in V} C_{v,j}^D \rightarrow V_j$ 
13:  end for
14:  for  $j \in J, v \in V$  do
15:    if  $(v, V_j) = (j, J_v)$  cannot be found then
16:      Remove  $(v, j)$  from set  $J_v$ 
17:    end if
18:  end for
19:  Renew  $C_{v,j}^D$  by  $J_v$ 
20: end while
21: Output  $J_v$ 

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* H denotes the set of channel gains $(h_v, h_{v,j}^D, h_{v,j}^C, h_{v,j}^p)$.

VI. SIMULATION RESULTS

The performance (total capacity of D2D pairs) of the RICE algorithm is compared with the exclusive spectrum allocation (ESA) strategy and adaptive subcarrier allocation (ASA) strategy. ESA uses a RB reusing strategy different from RICE and ASA. In ESA, different D2D pairs cannot reuse the same RB. ASA is a low complexity algorithm for D2D communications, that can provide a better performance in a system with a large number of D2D pairs. In ASA, D2D pairs, which have low channel gains between each other ($h_{v,j}^D$), can reuse the same RBs, and the transmission power is determined by a novel geometric water-filling algorithm [19]. The capacity is also comparable with the largest theoretical capacity of the system in the simulations.

In the simulations, the cell radius is set as 200 meters. The total bandwidth is 10 MHz, which includes 50 RBs in one timeslot. The transmission power of user equipments ($P_{v,j}^{D,max}$) is set as 20 dBm. The SINR threshold for D2D users is set as 4 dB. The noise power (N_v) is -30 dBm per RB [20], [21].

In order to provide different channel gains, the distance between users follows the Gaussian distribution with different mean values (μ) and different variances (σ^2). The distance between the BS and all the users, including cellular users and D2D users, follows the Gaussian distribution with $\mu_B = 100$, $\sigma_B^2 = 1000$. The distance between each un-paired D2D users follows the Gaussian distribution with $\mu_D = 50$, $\sigma_D^2 = 500$. The distance between two devices in one D2D pair follows the Gaussian distribution with $\mu_P = 20$, $\sigma_P^2 = 100$.

The capacity variation with different numbers of D2D pairs in one timeslot is shown in Fig. 2. The total capacity of D2D pairs calculated by RICE, ESA and ASA keeps increasing as more D2D pairs are involved, and the ASA case increases faster than the RICE case. The ASA case increases the fastest with a small number of D2D pairs, and the increasing rate decreases with the increasing number of D2D pairs. The total capacity calculated by RICE, ESA and ASA is normalized by the theoretical largest capacity of the system, as shown in Fig. 3. As more D2D pairs are added to the system that follows the same distribution, the normalized capacity calculated by the RICE algorithm decreases, while the counterparts of ESA and ASA keep rising. The curves of RICE and ASA are close to each other for a large number of D2D pairs. Although ASA increases faster when the number of D2D users is small, RICE still provides the larger capacity than ESA and ASA do.

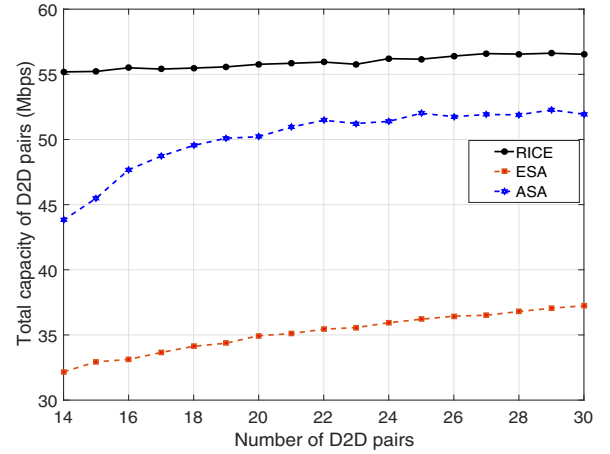


Fig. 2. Total capacity of D2D pairs.

Fig. 4 and 5 indicate the performance calculated by RICE and ESA with different average distance (μ_D) between D2D users. The number of D2D pairs in this simulation is 20. The average distance between D2D users indicates the density of users. A smaller μ_D means a higher density of D2D users. Fig. 4 shows that the total capacity of D2D pairs calculated by RICE and ESA does not vary much with μ_D . The capacity of ASA is sensitive to the distance between D2D users. The normalized capacity in Fig. 5 increases as the average distance between the D2D users increases for both RICE and ESA, implying that the largest theoretical capacity of the system decreases with the increasing density of D2D users. Even the the largest theoretical capacity of the system decreases with the distance between D2D users, the normalized capacity of

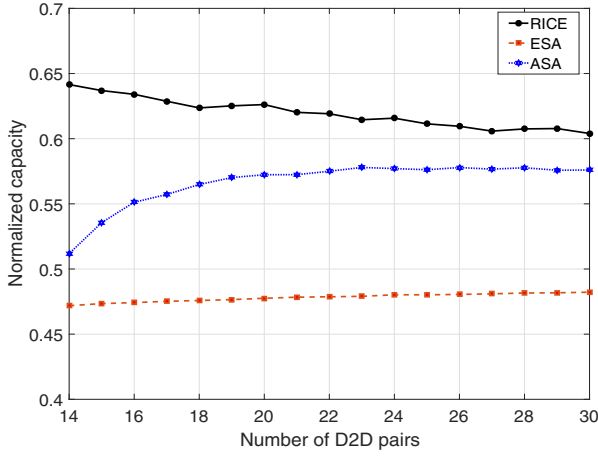


Fig. 3. Normalized capacity of D2D pairs.

the ASA case decreases as well. The curves of ASA and ESA are closer when the distance between D2D users is small. In a system with more D2D pairs, the capacity calculated by the RICE algorithm is closer to the theoretical largest capacity, while those of the ASA and ESA become close.

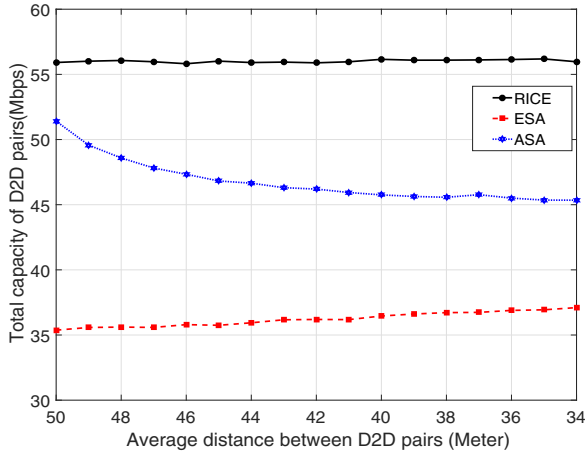


Fig. 4. Total capacity of D2D pairs.

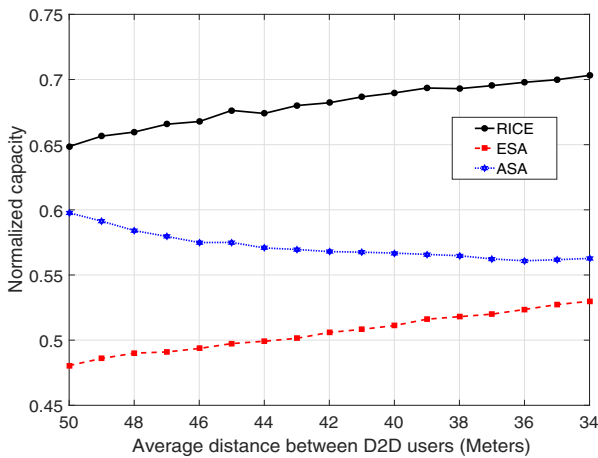


Fig. 5. Normalized capacity of D2D pairs.

VII. CONCLUSIONS

In this paper, we have proposed a system to provision both D2D users and cellular users, where multiple D2D users reuse the same downlink spectrum of cellular users. Based on the analysis of the interference of the system, we have proposed a heuristic algorithm, RICE, to maximize the capacity of D2D users. We have compared the capacity with those of the exclusive spectrum allocation (ESA) strategy and adaptive subcarrier allocation (ASA), and demonstrated that RICE can provide much higher capacity than ESA and ASA do. By analyzing the normalized capacity provided by RICE, ESA and ASA, the RICE algorithm is validated to be more suitable for a system with a high density of D2D users.

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